



SALES AND PRODUCT MANAGEMENT SYSTEM

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PROJECT OBJECTIVES



To create a structured database for managing sales data.



To ensure data integrity and prevent duplication.



To facilitate easy access to sales and product information for analysis.

DATABASE STRUCTURE

- Description of the four main tables:
 - **Geo:** Contains geographic information.
 - **People:** Stores salesperson details.
 - **Products:** Lists product information.
 - **Sales:** Records sales transactions.
- These tables are connected with foreign keys for efficient and robust database structure.

```
CREATE TABLE `sales` (  
  `SPID` varchar(6),  
  `GeoID` varchar(4),  
  `PID` varchar(6),  
  `SaleDate` datetime DEFAULT NULL,  
  `Amount` int DEFAULT NULL,  
  `Customers` int DEFAULT NULL,  
  `Boxes` int DEFAULT NULL,  
  foreign key(GeoID) references geo(GeoID),  
  foreign key(SPID) references people(SPID),  
  foreign key(PID) references products(PID)  
);
```

ENSURING DATA INTEGRITY: COLUMN MODIFICATIONS FOR CLARITY:

Explanation: RENAMES "GEO" COLUMN TO "COUNTRY" IN THE "GEO" TABLE.

Input

```
alter table geo rename column Geo to Country;
```

Output

	GeoID	Geo	Region
▶	G1	India	APAC
	G2	USA	Americas
	G3	Canada	Americas
	G4	New Zealand	APAC
	G5	Australia	APAC
	G6	UK	Europe
●	NULL	NULL	NULL

	GeoID	Country	Region
▶	G1	India	APAC
	G2	USA	Americas
	G3	Canada	Americas
	G4	New Zealand	APAC
	G5	Australia	APAC
	G6	UK	Europe
●	NULL	NULL	NULL

MODIFYING THE REGION COLUMN IN THE GEO TABLE TO DISALLOW NULL VALUES,
ENSURING ALL RECORDS HAVE VALID DATA.

Input

```
alter table geo modify column Region text not null;  
alter table geo modify column Country text not null;  
describe geo;
```

Output

	Field	Type	Null	Key	Default	Extra
►	GeoID	varchar(4)	NO	PRI	NONE	
	Country	text	NO		NONE	
	Region	text	NO		NONE	

AGGREGATE ANALYSIS OF SALES AND BOXES SOLD BY PRODUCT CATEGORY

Input

```
select  p.Category,  
        SUM(s.Amount) AS Total_Sales_Amount,  
        SUM(s.Boxes) AS Total_Boxes_Sold  
from sales s  
join products p on s.PID = p.PID  
GROUP BY p.Category;
```

Output

	Category	Total_Sales_Amount	Total_Boxes_Sold
▶	Bars	471835	27135
	Bites	314440	17893
	Other	134498	6082

TARGETED ANALYSIS OF SALES RECORDS BY SALESPERSON

Explanation: RETRIEVES SALES RECORDS FOR SPECIFIED SALESPERSONS USING THE IN OPERATOR.

Input

```
SELECT *  
FROM sales  
WHERE SPID IN ('SP01', 'SP02', 'SP07', 'SP08');
```

Output

	SPID	GeoID	PID	SaleDate	Amount	Customers	Boxes
▶	SP01	G4	P04	2021-01-01 00:00:00	8414	276	495
	SP01	G4	P15	2021-01-01 00:00:00	259	32	22
	SP01	G5	P01	2021-01-11 00:00:00	7238	67	315
	SP01	G3	P19	2021-01-12 00:00:00	490	188	35
	SP01	G2	P14	2021-01-18 00:00:00	2331	38	167
	SP02	G3	P14	2021-01-01 00:00:00	532	317	54
	SP02	G4	P13	2021-01-01 00:00:00	2506	99	148
	SP02	G1	P05	2021-01-08 00:00:00	2814	69	149
	SP02	G3	P15	2021-01-12 00:00:00	2653	147	332
	SP02	G2	P12	2021-01-14 00:00:00	14532	142	969
	SP02	G4	P15	2021-01-14 00:00:00	4011	260	335
	SP02	G4	P05	2021-01-15 00:00:00	3430	120	202
	SP07	G4	P11	2021-01-05 00:00:00	7399	420	275
	SP07	G4	P04	2021-01-05 00:00:00	4865	168	271
	SP07	G2	P15	2021-01-06 00:00:00	1470	57	184
	SP07	G4	P13	2021-01-13 00:00:00	2121	130	89
	SP07	G6	P10	2021-01-19 00:00:00	8218	205	294

FILTERING OF CATEGORIES PRODUCTS BY USING NAMES WITH WILDCARD OPERATOR:

EXPLANATION: RETRIEVES PRODUCTS WHOSE CATEGORY NAMES START WITH 'B' USING THE LIKE OPERATOR.

Input

```
select PID, Product, Category
from Products
where Category like 'B%';
```

Output

	PID	Product	Category
►	P01	Milk Bars	Bars
	P02	50% Dark Bites	Bites
	P03	Almond Choco	Bars
	P04	Raspberry Choco	Bars
	P05	Mint Chip Choco	Bars
	P06	Eclairs	Bites
	P08	99% Dark & Pure	Bars
	P09	Orange Choco	Bars
	P10	Spicy Special Slims	Bites
	P11	After Nines	Bites
	P12	Fruit & Nut Bars	Bars
	P13	85% Dark Bars	Bars

IDENTIFYING HIGH-VALUE SALES TRANSACTIONS

Explanation:RETRIEVES THE TOP 10 SALES RECORDS ORDERED BY AMOUNT USING THE LIMIT CLAUSE.

Input

```
select SPID, GeoID, PID, SaleDate,
       Amount, Customers, Boxes
from sales
order by Amount desc limit 10;
```

Output

	SPID	GeoID	PID	SaleDate	Amount	Customers	Boxes
►	SP18	G2	P21	2021-01-04 00:00:00	19229	64	1013
	SP24	G2	P11	2021-01-14 00:00:00	18704	78	585
	SP23	G1	P16	2021-01-05 00:00:00	17248	163	664
	SP13	G2	P20	2021-01-19 00:00:00	16380	203	1092
	SP05	G6	P20	2021-01-14 00:00:00	15785	209	1128
	SP10	G1	P06	2021-01-01 00:00:00	15596	32	975
	SP08	G1	P05	2021-01-19 00:00:00	15407	103	771
	SP20	G5	P12	2021-01-14 00:00:00	14574	86	810
	SP02	G2	P12	2021-01-14 00:00:00	14532	142	969
	SP25	G6	P05	2021-01-01 00:00:00	14273	335	752

TOTAL SALESPERSON COUNT BY TEAM:

Explanation: This query calculates total number of salespersons in each team from the 'people' table

Input

```
select Team, count(Salesperson)
from people
group by Team;
```

Output

	Team	count(Salesperson)
►	Yummies	9
	Delish	10
	Unassigned	5
	Jucies	9

IDENTIFYING HIGH-PERFORMING PRODUCTS BY SALES VOLUME

Explanation: FILTERS PRODUCTS WITH TOTAL SALES GREATER THAN \$20,000 USING GROUP BY AND HAVING CLAUSES.

Input

```
select SPID, sum(Amount) as TotalSales
FROM sales group by SPID
having sum(Amount) > 20000;
```

Output

	SPID	TotalSales
▶	SP02	30478
	SP03	35238
	SP04	44639
	SP05	45472
	SP06	46690
	SP07	24073
	SP08	63749
	SP09	25914
	SP10	54208
	SP11	68509
	SP12	31731
	SP13	45591

FINDING THE PRODUCT WITH THE HIGHEST COST PER BOX.

Explanation:QUERY RETRIEVES THE RECORD OF PRODUCT HAVING HIGHEST COST_PER_BOX

Input

```
select Product, Cost_per_box
from products
where Cost_per_box = (
    select max(Cost_per_box)
    from products);
```

Output

	Product	Cost_per_box
▶	Almond Choco	9.6

RETRIEVING SALES RECORDS FOR THE SALESPERSON "BARR FAUGHNY" IN THE "BARS" CATEGORY, ORDERED BY SALES AMOUNT

Input

```
select p.Salesperson, p.Team, p.SPID, s.SPID, s.Amount,
       s.Customers, pr.Category, pr.Product, pr.PID
from sales s
join people p on s.SPID = p.SPID
join products pr on s.PID = pr.PID
where pr.category = "BARS" and p.Salesperson = "Barr Faughny"
order by s.Amount desc;
```

Output

	Salesperson	Team	SPID	SPID	Amount	Customers	Category	Product	PID
▶	Barr Faughny	Yummies	SP01	SP01	8414	276	Bars	Raspberry Choco	P04
	Barr Faughny	Yummies	SP01	SP01	7238	67	Bars	Milk Bars	P01
	Barr Faughny	Yummies	SP01	SP01	259	32	Bars	Baker's Choco Chips	P15

STREAMLINED DATA ENTRY: ADDING A NEW SALESPERSON VIA STORED PROCEDURE

Explanation: CREATES A STORED PROCEDURE TO INSERT A NEW SALESPERSON INTO THE "PEOPLE" TABLE, ENHANCING DATA MANAGEMENT EFFICIENCY.

Input

```
delimiter $$
create procedure insert_new_salesper(
  in p_Salesperson text,
  in p_SPID varchar(6),
  in p_Team text,
  in p_Location text
)
begin
  insert into people(Salesperson, SPID, Team, Location)
  values(p_Salesperson, p_SPID, p_Team, p_Location);
end$$
delimiter ;

CALL insert_new_salesper('Angad', 'SP34', 'Mango', 'Ratnagiri');
```

Output

	Salesperson	SPID	Team	Location
	Mallorie Waber	SP21		Seattle
	Jehu Rudeforth	SP22		Seattle
	Van Tuxwell	SP23	Yummies	Seattle
	Roddy Speechley	SP24	Delish	Seattle
	Camilla Castle	SP25	Delish	Seattle
	Janene Hairsine	SP26	Delish	Paris
	Niall Selesnick	SP27	Jucies	Paris
	Ebonee Roxburgh	SP28		Paris
	Zach Polon	SP29	Yummies	Paris
	Orton Livick	SP30	Yummies	Paris
	Gray Seamon	SP31	Delish	Paris
	Benny Karolovsky	SP32	Jucies	Paris
	Dyna Doucette	SP33	Jucies	Paris
	Angad	SP34	Mango	Ratnagiri

EFFICIENT DATA ANALYSIS: AVERAGE SALES CALCULATION BY REGION BY CREATING A STORED FUNCTION

EXPLANATION: DEFINES A FUNCTION TO CALCULATE AVERAGE SALES FOR A GIVEN REGION, IMPROVING DATA ANALYSIS EFFICIENCY.

Input

```
delimiter $$
create function avg_sales_by_region(
e_Region text)
returns decimal(10,2)
deterministic
begin
    declare averageAmount decimal(10,2);
    select avg(s.Amount) into averageAmount
    from sales s join geo g
    on s.GeoID = g.GeoID
    where g.Region = e_Region;
    return averageAmount;
end$$
delimiter ;
select avg_sales_by_region("Americas");
```

Output

	avg_sales_by_region("Americas")
▶	6045.18

ENSURING DATA INTEGRITY: TRIGGER FOR DUPLICATE SALESPERSON ID PREVENTION

Explanation: CREATES A TRIGGER TO PREVENT DUPLICATE SALESPERSON IDS IN THE PEOPLE TABLE, ENSURING DATA INTEGRITY DURING INSERTIONS.

Input

```
delimiter $$
create trigger prevent_duplicate_spid
before insert on people
for each row
begin
    declare existing_count int;
    select count(*) into existing_count
    from people
    where SPID = new.SPID;

    if existing_count > 0 then
        signal sqlstate '45000'
        set message_text = 'Duplicate SPID is not allowed';
    end if;
end$$
delimiter ;
insert into people values("Angad", "SP01", "Mango", "Ratnagiri");
```

Output

Error Code: 1644. Duplicate SPID is not allowed

COMPREHENSIVE VIEW OF PRODUCTS, SALES, AND GEOGRAPHY

EXPLANATION: CREATES A VIEW TO DISPLAY DETAILED PRODUCT AND SALES INFORMATION, STREAMLINING DATA RETRIEVAL FOR ANALYSIS.

Input

```
create view show_product_info as
select g.Country, p.Salesperson,
       pr.Product, pr.Category,
       pr.Size, s.SaleDate,
       s.Amount, s.Boxes
from
geo g join sales s on s.GeoID = g.GeoID join
people p on p.SPID = s.SPID join
products pr on pr.PID = s.PID;

select * from show_product_info;
```

Output

	Country	Salesperson	Product	Category	Size	SaleDate	Amount	Boxes
▶	India	Rafaelita Blaksland	85% Dark Bars	Bars	SMALL	2021-01-01 00:00:00	2184	122
	India	Brien Boise	Eclairs	Bites	LARGE	2021-01-01 00:00:00	15596	975
	India	Curtice Advani	After Nines	Bites	LARGE	2021-01-01 00:00:00	8561	330
	India	Van Tuxwell	Organic Choco Syrup	Other	SMALL	2021-01-05 00:00:00	17248	664
	India	Jan Morforth	Orange Choco	Bars	LARGE	2021-01-05 00:00:00	3059	279
	India	Wilone O'Kielt	Smooth Sliky Salty	Bars	SMALL	2021-01-06 00:00:00	644	34
	India	Ches Bonnell	After Nines	Bites	LARGE	2021-01-06 00:00:00	4935	171
	India	Madelene Upcott	Manuka Honey Choco	Other	SMALL	2021-01-07 00:00:00	11228	388
	India	Dennison Crosswaite	Mint Chip Choco	Bars	LARGE	2021-01-08 00:00:00	2814	149
	India	Karlen McCaffrey	85% Dark Bars	Bars	SMALL	2021-01-08 00:00:00	9562	479
	India	Ches Bonnell	Caramel Stuffed Bars	Bars	SMALL	2021-01-08 00:00:00	2807	123
	India	Curtice Advani	White Choc	Other	SMALL	2021-01-12 00:00:00	6776	424
	India	Dotty Strutley	Spicy Special Slims	Bites	LARGE	2021-01-12 00:00:00	8610	345

The background of the slide features a collage of financial data visualizations. It includes a bar chart at the top left, a line graph with a shaded area in the middle, and a pie chart at the bottom right. Overlaid on these are various numerical values and text fragments, such as '3457.71', 'Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', and '300.40'. The overall color scheme is a mix of blue, orange, and white, giving it a professional, data-driven appearance.

CONCLUSION

In this project, I utilized SQL to manage and analyze sales data through key features such as joins for comprehensive reporting, stored procedures for inserting new salespersons, and functions to calculate average sales by region. Additionally, I implemented triggers to ensure data integrity by preventing duplicate entries. This approach demonstrated my ability to leverage database functionalities to drive informed business decisions and enhance operational efficiency.



THANK YOU